

Problem 1. Solve the radial wave equation on a circular membrane with $c = 1$, $R = 1$, zero initial velocity, and initial displacement $u(r, 0) = f(r)$, where

$$f(r) = 1.108J_0(2.4048r) - 0.140J_0(5.5201r) + 0.045J_0(8.6537r).$$

Additionally, find all the harmonic frequencies and the radius of each of the (inner) nodal circle for each mode (m) of the solution.

For reference:

$$\alpha_1 = 2.4048, \quad \alpha_2 = 5.5201, \quad \alpha_3 = 8.6537, \quad \alpha_4 = 11.7915, \quad \alpha_5 = 14.9309$$

[10 marks]

Solution. Note that the form for $f(r)$ is

$$f(r) = A_1J_0(\alpha_1r/R) + A_2J_0(\alpha_2r/R) + A_3J_0(\alpha_3r/R),$$

where $R = 1$. Therefore, in the solution of $u(r, t)$, we would have

$$A_1 = 1.108, \quad A_2 = -0.140, \quad A_3 = 0.045, \quad A_m = 0 \text{ for } m > 3.$$

Also, as $c = R = 1$, then $\lambda_m = ck_m = \frac{c\alpha_m}{R} = \alpha_m$. Thus we get the solution as

$$u(r, t) = 1.108 \cos(\alpha_1 t) J_0(\alpha_1 r) - 0.140 \cos(\alpha_2 t) J_0(\alpha_2 r) + 0.045 \cos(\alpha_3 t) J_0(\alpha_3 r).$$

The harmonic frequencies (in Hz) are given by $f_m = \frac{\lambda_m}{2\pi}$, so

$$\begin{aligned} f_1 &= \frac{2.4048}{2\pi} = 0.35663 \text{ Hz} \\ f_2 &= \frac{5.5201}{2\pi} = 0.87855 \text{ Hz} \\ f_3 &= \frac{8.6537}{2\pi} = 1.377279 \text{ Hz.} \end{aligned}$$

Next, for mode $m = 1$, there is no inner nodal circle. For mode $m = 2$, there is one inner nodal circle at radius $r_1 = \frac{\alpha_1}{\alpha_2} R \approx 0.4356$.

Finally, for mode $m = 3$, there are two inner nodal circles at

$$r_1 = \frac{\alpha_1}{\alpha_3} R \approx 0.2778, \quad r_2 = \frac{\alpha_2}{\alpha_3} R \approx 0.6379.$$

□